

Energy Meter



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Electromechanical energy meters have been the standard for metering the electricity since billing began. But these are now being gradually replaced by electronic digital energy meters. Presented here is a simple energy meter using Analog Device's ADE7757 chip for single-phase, 2-wire (phase and neutral) systems used in households. IC ADE7757 is a low-cost, single-phase solution for electrical energy measurement. You can use this solution even for individual appliances to see how much energy they are consuming.

Its salient features are:

1. Can read up to 999999 units (kWh) with a resolution of 0.01 units
2. Designed for normal 230V AC and maximum line current of 30 amps
3. The meter count is 100 pulses/kWh, i.e., 100 pulses will be required to register one unit

Circuit and working

Fig. 1 shows the circuit diagram of the energy meter which is built around energy-metering IC with integrated oscillator ADE7757 (IC1), microcontroller AT89C52 (IC2), EEPROM AT24C02 (IC3), 5V voltage regulator 7805 (IC5),

TABLE I
Test Points

Test point	Details
TP0	0V, AGND
TP1	5V (w.r.t. TP0)
TP2	Train of pulses of frequency 32 times of TP3
TP3	100 pulses/kWh
TP4	100 pulses/kWh
TP5	0V, GND
TP6	5V (w.r.t. TP5)

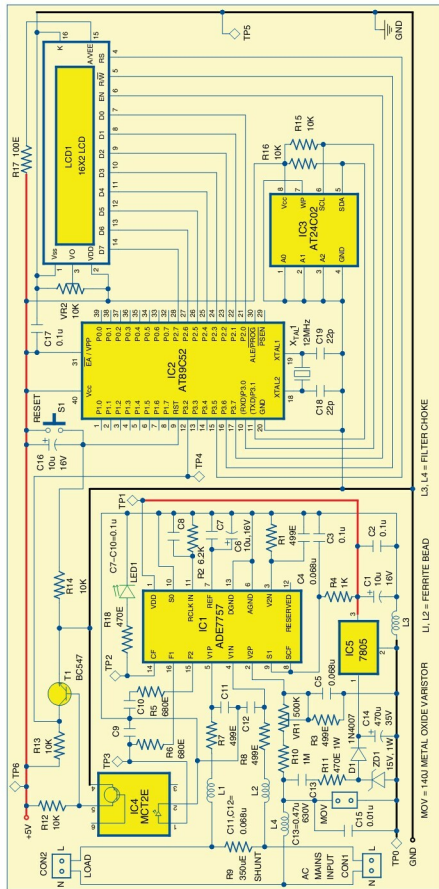


Fig. 1: Circuit diagram of the energy meter